

SYDNEY TECHNICAL HIGH SCHOOL



MATHEMATICS DEPARTMENT

YEAR 11 EXTENSION 1

H.S.C. ASSESSMENT TASK 1, DECEMBER 2011

Name: _____ Teacher: _____

- Time allowed: 70 minutes.
- Start each question on a new page.
- Diagrams are not to scale.
- Show necessary working.
- Full marks may not be awarded for poorly arranged work or illegible writing.

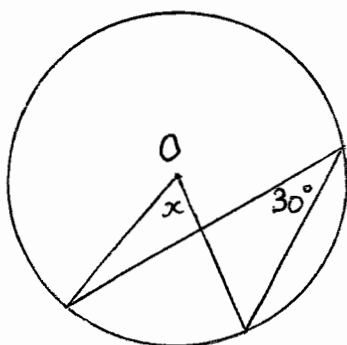
Question 1	Question 2	Question 3	Question 4	Question 5	TOTAL
/10	/11	/9	/10	/10	/50

Question 1

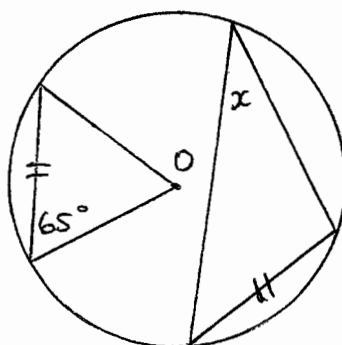
- a) Find the value of each pronumeral. Reasons are not required. O is the centre of each circle and diagrams are not to scale.

5

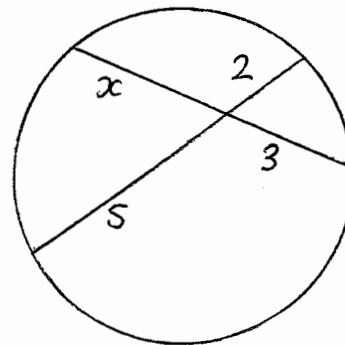
i)



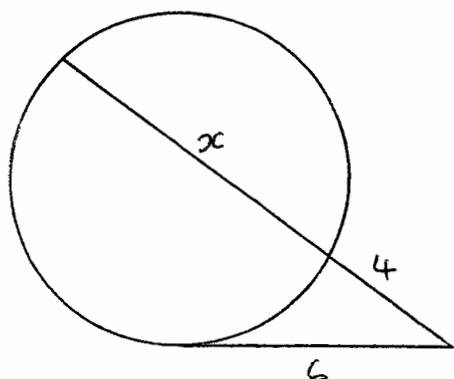
ii)



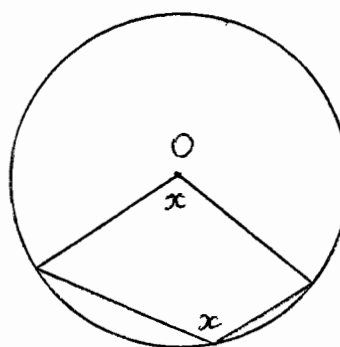
iii)



iv)



v)



- b) A sequence is given by $T_n = \frac{n-1}{n}$.

i) Which term of the sequence is 0.99?

1

ii) Simplify $T_{n+1} : T_n$

1

c) Evaluate $\sum_{n=20}^{100} (2n - 4)$

2

d) Find the equation of the chord of contact from $(-1, -2)$ to the parabola $x^2 = 4y$

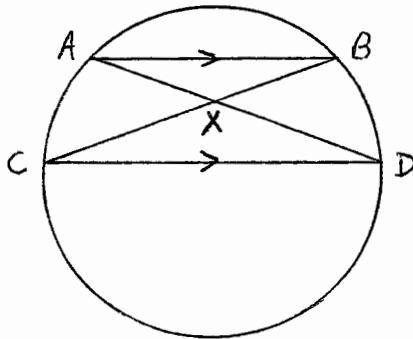
1

Question 2 (start a new page)

a) For a certain series, the sum to n terms is $S_n = n^2 - 4n$. Find:

- i) the seventh term. 1
- ii) the n th term in simplest form. 2

b)



AB and CD are parallel chords. AD and CB intersect at X .

Prove that $\triangle CXD$ is isosceles.

2

c) The parabola $x = 4t$, $y = 2t^2$ has points P and Q with parameters " p " and " q ".

- i) Find the equation of the chord PQ . 2
- ii) If PQ is a focal chord, show that $pq = -1$ 1
- iii) Find the coordinates of M , the midpoint of PQ . 1
- iv) Show that the locus of M is the parabola $x^2 = 4y - 8$. 2

Question 3 (start a new page)

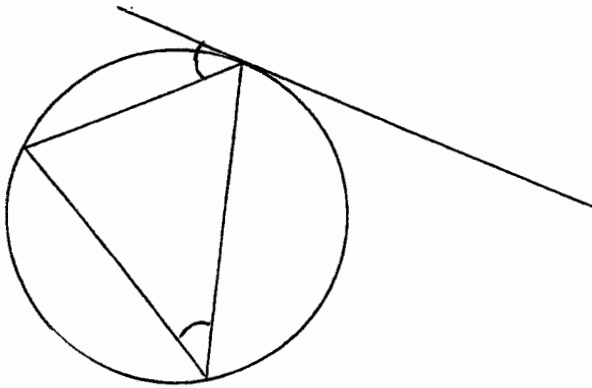
a) Rewrite $3 + 5 + 7 + \dots + 99$ using sigma notation, starting with $n = 1$.

1

b) Find the sum to 30 terms of the sequence $T_n = 2 + 2^n - 2n$.

3

c)



Write the fully worded property that applies to the marked angles above.

1

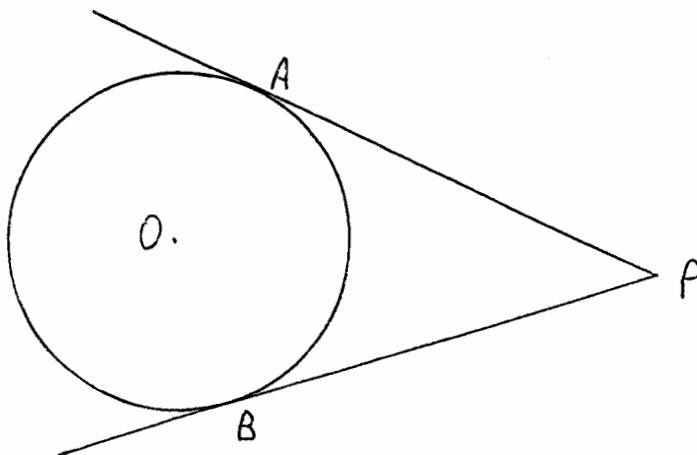
d) Prove by Mathematical Induction that the sum of the first n terms of a geometric series

$a + ar + ar^2 + \dots + ar^{n-1}$, is $S_n = \frac{a(r^n - 1)}{r - 1}$ for positive integers n ($r \neq 1$).

4

Question 4 (start a new page)

a)

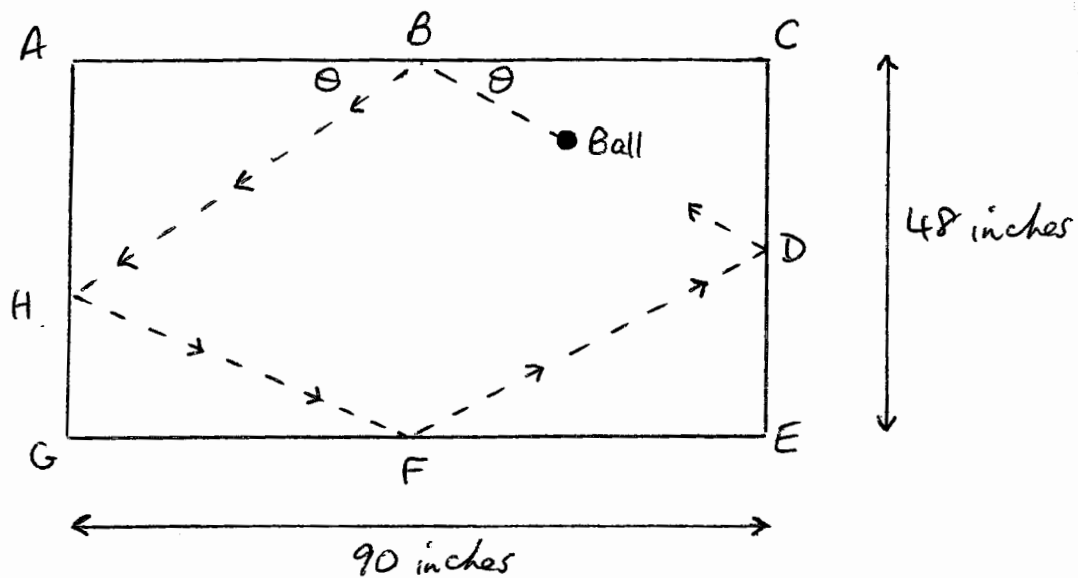


Two tangents are drawn to a circle, centre O , from an external point P to touch the circle at A and B .

Prove that the tangents are equal in length.

2

b)



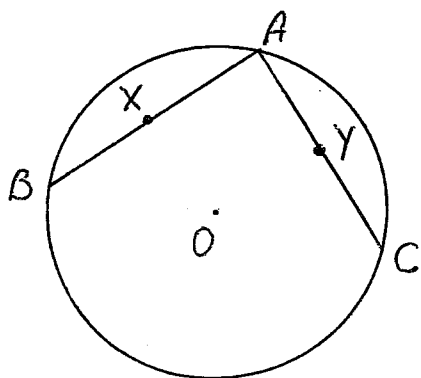
A rectangular pool table is 90 inches long and 48 inches wide (it's American!).

When hit, the ball shown makes angles of θ on the first rebound at B and continues rebounding perfectly (equal angles) off each side, returning to its starting position.

- i) Copy the diagram neatly and mark all angles in terms of θ . 1
- ii) Why is $BHFD$ a parallelogram? 1
- iii) Which congruence test is used to prove that $\triangle BCD \equiv \triangle FGH$? (do not prove congruence) 1
- iv) Let $BC = m$, $CD = n$. Show that $m:n = 15:8$ 3
- v) Find the perimeter of parallelogram $BHFD$. 2

Question 5 (start a new page)

a)



AB and AC are chords of a circle, centre O . X and Y are midpoints of AB and AC .

- i) Prove that A, X, O, Y form a cyclic quadrilateral.
- ii) Describe where the centre of the circle $AXOY$ is.

3

1

- b) A man borrows \$5000 from the bank at a reducible interest rate of 12% p.a. He repays \$400 per month.

Let A_n represent the amount still owing on the loan after n months.

- i) Write an expression for A_1 and show that $A_2 = 5000 \times 1.01^2 - 400(1.01 + 1)$.
- ii) Show that $A_n = 5000 \times 1.01^n - 40000(1.01^n - 1)$.
- iii) Hence show that the number of months, n , could be found using $1.01^n = \frac{8}{7}$

2

2

2

END OF TEST

SOLUTIONS

① a) i) 60° ii) 25° iii) $3x = 10$ iv) $\frac{x}{2} + x = 180^\circ$ v) $4(x+4) = 36$
 $x = 3\frac{1}{3}$ $3x = 360$ $x = 5$
 $x = 120$

b) i) $\frac{n-1}{n} = 0.99$ ii) $\frac{n}{n+1} : \frac{n-1}{n} = \frac{n}{n+1} \times \frac{n}{n-1}$
 $n-1 = 0.99n$
 $0.01n = 1$
 $n = 100$
 $= \frac{n^2}{n^2-1}$

c) $S_{81} = \frac{81}{2} (36 + 196)$
 $= 9396$

d) $-x = 2(y+2)$
 $\therefore x + 2y + 4 = 0$

② a) i) $T_7 = S_7 - S_6$ ii) $T_n = S_n - S_{n-1}$
 $= (49 - 28) - (36 - 24)$
 $= 21 - 12$
 $= 9$
 $= n^2 - 4n - [(n-1)^2 - 4(n-1)]$
 $= n^2 - 4n - (n^2 - 2n + 1 - 4n + 4)$
 $= 2n - 5$

b) $\angle A = \angle D$ (alternate angles, parallel lines)
 $\angle A = \angle C$ (equal angles on same arc BD)
 $\therefore \angle C = \angle D$
 $\therefore \triangle CxD$ is isosceles (equal base angles)

c) i) $P(4p, 2p^2) \quad Q(4q, 2q^2)$

$$\begin{aligned} \text{chord } PQ: \frac{y-2p^2}{x-4p} &= \frac{2q^2-2p^2}{4q-4p} \\ &= \frac{2(\cancel{q-p})(q+p)}{4(\cancel{q-p})} \\ &= \frac{p+q}{2} \end{aligned}$$

$$\therefore 2y-4p^2 = (p+q)(x-4p)$$

ii) Subst. $(0, 2)$

$$4-4p^2 = (p+q)(-4p)$$

$$4-4p^2 = -4p^2 - 4pq$$

$$-4pq = 4$$

$$pq = -1 \text{ as reqd.}$$

iii) $M\left(\frac{4p+4q}{2}, \frac{2p^2+2q^2}{2}\right)$

$$= M(2p+2q, p^2+q^2)$$

iv) $x = 2p+2q$
 $= 2(p+q)$

$$\begin{aligned} \therefore p+q &= \frac{x}{2}, \quad y = p^2+q^2 \\ &= (p+q)^2 - 2pq \\ &= \left(\frac{x}{2}\right)^2 + 2 \\ &= \frac{x^2}{4} + 2 \end{aligned}$$

$$\therefore 4y = x^2 + 8 \quad \text{or } x^2 = 4y - 8 \text{ as reqd.}$$

$$(3) a) \sum_{n=1}^{49} (2n+1)$$

$$b) \text{sum G.P. } (2^n) + \text{sum A.P. } (-2n+2)$$

$$= \frac{2(2^{30}-1)}{2-1} + \frac{30}{2}(0-58)$$

$$= 2147483646 - 870$$

$$= 2,147,482,776$$

c) The angle between a tangent and a chord, at the point of contact, is equal to the angle in the alternate segment standing on the same arc/chord

d) Prove true for $n=1$

$$\begin{aligned} \text{LHS} &= a, \text{ RHS} = \frac{a(r^1 - 1)}{r - 1} \\ &= a \\ &= \text{LHS} \end{aligned}$$

Assume true for $n=k$, i.e. assume $S_k = \frac{a(r^k - 1)}{r - 1}$

Prove true for $n=k+1$, i.e. prove that $S_{k+1} = \frac{a(r^{k+1} - 1)}{r - 1}$

$$\text{Use } S_{k+1} = S_k + T_{k+1}$$

$$= \frac{a(r^k - 1)}{r - 1} + ar^k$$

$$= \frac{a(r^k - 1) + ar^k(r - 1)}{r - 1}$$

$$= \frac{\cancel{ar^k} - a + ar^{k+1} - \cancel{ar^k}}{r - 1}$$

$$= \frac{a(-1 + r^{k+1})}{r - 1}$$

$$= \frac{a(r^{k+1} - 1)}{r - 1} \text{ as reqd.}$$

$\therefore$ if the result is true for $n=k$, then it has been proven true for $n=k+1$

The result is true for $n=1$, and from above it must be true for $n=1+1=2$, then $n=2+1=3$ and so on for all pos. integral n .

Question 4.

a) $OA = OB$ (equal radii)

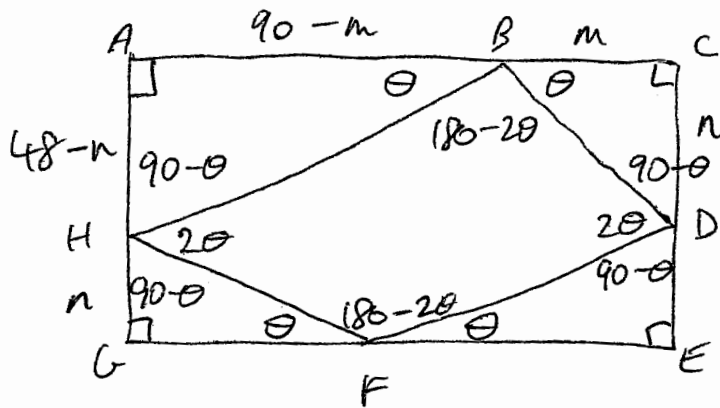
OP is common

$\angle OAP = \angle OBP = 90^\circ$ (radius $\perp$ tangent)

$\therefore \triangle OAP \equiv \triangle OBP$ (RHS)

$\therefore AP = BP$ (corresponding sides in congruent triangles)

b)



i) above

ii) opposite angles are equal

iii) AAS

iv) Since congruent, then $HG = n$
(corresponding sides)

$\therefore AH = 48 - n$

Now $\triangle ABH \parallel \triangle BCD$ (equiangular from i)

$\therefore \frac{m}{n} = \frac{90-m}{48-n}$ (equal ratio corresp. sides)

$\therefore 48m - mn = 90 - mn$

$\therefore \frac{m}{n} = \frac{90}{48}$

$\therefore m : n = 15 : 8$

v) Let $BC = 15$, $CD = 8$
 $\therefore BD = 17$ (Pythagora)

Also, $AB = 75$, $AH = 40$

$\therefore BH = 85$ (Pythag -)

$\therefore$ perimeter $BHFD$
 $= 2 \times 17 + 2 \times 85$
 $= 204$ inches.

⑤ a) i) $\angle OXA = 90^\circ$ (centre to midpoint of chord $\perp$ chord)

Similarly $\angle OYA = 90^\circ$

$\therefore AXOY$ is a cyclic quadrilateral (opposite angles supplementary)

ii) centre is midpoint of OA .

b) i) $A_1 = 5000 \times 1.01 - 400$

$A_2 = A_1 \times 1.01 - 400$
 $= 5000 \times 1.01^2 - 400 \times 1.01 - 400$
 $= \underline{5000 \times 1.01^2 - 400(1.01 + 1)}$

ii) $A_n = 5000 \times 1.01^n - 400 \underbrace{(1.01^{n-1} + 1.01^{n-2} + \dots + 1)}_{S_n}$
 $S_n = \frac{1(1.01^n - 1)}{1.01 - 1}$

$\therefore A_n = 5000 \times 1.01^n - \frac{400(1.01^n - 1)}{0.01}$
 $= 5000 \times 1.01^n - 40000(1.01^n - 1)$ as reqd.

iii) $A_n = 0$ at end of loan

$\therefore 40000(1.01^n - 1) = 5000 \times 1.01^n$

$40000 \times 1.01^n - 40000 = 5000 \times 1.01^n$

$\therefore 35000 \times 1.01^n = 40000$

$\therefore 1.01^n = \frac{40000}{35000}$

$= \frac{8}{7}$ as reqd.